

ISRAEL VALLEJO MUNOZ

Marine Biologist | MSc
Environmental Data Analyst
GIS & Remote Sensing

CONTACT

Mexico City, Mexico
+52 55 2771 8407
israelvallejo.mz@gmail.com
LinkedIn
Portfolio

TECHNICAL SKILLS

GIS & Remote Sensing
QGIS 3.x · Google Earth Engine
Sentinel-2 · Landsat-8 · SAR

Programming & Data
R / RStudio · tidyverse · terra
sf · ggplot2 · Python · Git

Ecological Modelling
MaxEnt · ENMeval · MaxNet
CPUE · Surplus Production Models

GHG Accounting
GHG Protocol · Scopes 1-2-3
IPCC AR6 Emission Factors

Data Sources
Global Fishing Watch · GBIF
Bio-Oracle · WDPA

LANGUAGES

Spanish Native / C2
English Proficient / C1
Italian Intermediate / B2

COURSES & CERTIFICATIONS

- GHG Protocol: Foundations
GHG Protocol, 2026
- GIS Applied to Natural
Resources & AI Integration
- NADD Scientific Diver
- PADI Open Water Diver

Marine Biologist (MSc) · Environmental & Spatial Data Analyst
R · QGIS · Google Earth Engine · GHG Protocol · Remote Sensing · Ecological Modelling

PROFESSIONAL SUMMARY

Marine Biologist (MSc) and Environmental Data Analyst with 7+ years of combined research and analytical experience. Specialized in GIS, remote sensing, and species distribution modelling using R, QGIS, and Google Earth Engine. Trained in GHG Protocol (Scopes 1-2-3) and IPCC AR6 emission factor methodologies, with applied experience producing GHG inventories for the fisheries sector. Proven ability to translate complex environmental datasets into evidence-based insights for conservation, fisheries management, and sustainability reporting. Open to Sustainability Analyst, Environmental Data Analyst, and GIS/Remote Sensing roles across industries.

PROFESSIONAL EXPERIENCE

Fisheries Data Analyst (External Collaboration) Jan 2026 – Present

Sea Shepherd Global Remote

- Analyze fisheries enforcement and catch composition datasets across 2 high-priority regions (Mediterranean Sea and African coast), supporting conservation monitoring and campaign strategy.
- Conduct CPUE estimation and effort-standardized analyses to quantify fishing pressure and stock status, translating results into actionable intelligence for international scientific teams.
- Develop spatial visualizations of fishing effort and enforcement metrics in QGIS and GEE, directly informing patrol deployment and strategic reporting.

Environmental Data Analyst Jan 2025 – Present

Independent Consultant Remote

- Develop species distribution models (MaxEnt/ENMeval) and remote sensing analyses using R, QGIS, and Google Earth Engine, integrating multi-source spatial data (GBIF, Bio-Oracle, Sentinel-2, Landsat-8, Global Fishing Watch) for conservation and fisheries applications.
- Apply GHG Protocol Corporate Standard (Scopes 1-2-3) and IPCC AR6 emission factors to produce GHG inventories; recent project quantified ~805 kt CO2e for the artisanal fishing fleet of the Mexican Pacific across fuel combustion, electricity, and value chain emissions.
- Build fisheries stock assessment models (surplus production, CPUE standardization) and spatial analyses of fishing effort in Mexican MPAs and the Eastern Pacific Ocean.

Research Intern & Teaching Assistant Feb – Oct 2024

Universita degli Studi di Napoli Federico II Naples, Italy

- Processed tissue samples using spectrophotometric, chromatographic, and histological techniques to assess antioxidant profiles and reproductive physiology in aquaculture and wild fish.
- Teaching assistant for 6 months of undergraduate marine biology and aquaculture lab instruction, guiding experiments and evaluating scientific reports.

Research Assistant — Shark Ecology Jul – Aug 2018

Shark Research and Conservation Program Miami, USA

- Captured, tagged, and sampled 15+ large coastal sharks across 6-7 species, collecting blood, biopsy, and morphometric data for ecological research.

Research Assistant — Ichthyology 2016 – 2018

National Fish Collection & ICMYL (UNAM) Mexico City

- Processed and identified elasmobranch and demersal fish specimens; extracted otoliths for age and growth studies; contributed to taxonomic curation of the national ichthyological collection.

EDUCATION

MSc in Marine Biology and Aquaculture Oct 2024

Universita degli Studi di Napoli Federico II Naples, Italy

BSc in Biology Jan 2021

Universidad Nacional Autonoma de Mexico (UNAM) Mexico City